OPSIS®

Power Supply Unit PS150

User´s Guide

Limited warranty

The warranty conditions of this product are described in the purchase agreement between the buyer and the seller. However, a condition for any warranty is that the product is installed, operated and maintained in accordance with the specifications and instructions issued by Opsis AB including but not limited to this manual. This warranty is limited to replacement of defective material or documentation supplied by Opsis AB and in no event shall Opsis AB or its suppliers be liable for any loss of profit or any other commercial damage, including but not limited to direct or indirect, special, incidental, contingent or consequential damages resulting from any defects on the product or its documentation including damages from downtime, loss of goodwill, damage to or replacement of equipment or property.

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The equipment 1

The PS150 model is a switched power supply unit designed for the operation of the xenon lamps which are the light sources in the Opsis emitters. The xenon lamp wattage can be either 150 or 75.

The xenon lamp is automatically ignited at startup with a high voltage pulse. The pulse is generated by an ignition unit which is mounted on the emitter. After a few seconds the discharge in the lamp is stabilized and the lamp current is kept at a constant level.

The current from the power supply to the emitter is fed through a shielded cable. The normal output ratings are approximately 19 V/8 A for a 150 watts lamp, and 14 V/5 A for a 75 watts lamp.

The mains input can be either 115 or 230 V_{AC}.

The power supply is short-circuit proof. Maximum output current is limited to 8 A.

1.1 Specifications

Technical specifications: PS150 Power unit	
Input voltage	230 V_{AC} (+6 %, -10 %)
input current	0.6 A (normally)
fuses	2 x 4 A/250 V slow
Input voltage	115 V_{AC} (± 10 %)
input current	1.3 A (normally)
fuses	2 x 4 A/250 V slow
Power consumption	approx 220 W
Output power	150 W
output current	approx. 8 A_{DC}
output voltage	approx. 19 V_{DC}
Output power	75 W
output current	approx. 5 A_{DC}
output voltage	approx. 14 V_{DC}
Current stability	ripple max. ± 0.1 %
Output voltage when igniting lamp	80 V_{DC}
Working temperature range	-20 °C to 40 °C (-4 °F to 104 °F)
Degree of protection	IP65
Casing material	Aluminium
Dimensions (L x W x H)	280 x 115 x 270 mm^3 (11 x 4.5 x 10.6 in^3)
Weight	4.5 kg (10 lbs.)
Mains power cable length	2 m (6 ft. 6 in)

Technical specifications: PS150 Ignition unit	
High voltage pulse energy	max. 0.1 Ws
High voltage pulse frequency	approx. 0.2 s^{-1}
Casing material	Aluminium
Dimensions	174 x 98 x 75 mm^3 (6.8 x 3.8 x 2.9 in^3)
Cable (ignition unit - power unit)	3 x 1.5 mm^2
outer diameter	10.5 mm
cable length (standard)	5 m (15 ft)
cable length (max.)	100 m (300 ft)
connectors	AMP

1.2 Function

The power supply consists of a power supply card, a control card and an ignition unit. The input voltage to the power supply card should be either 115 or 230 V_{AC}. A jumper on the card should be set in accordance with the conditions. When no lamp current is flowing, the output voltage is 80 V_{DC}. This voltage supports the ignition of the lamp, and activates the ignition unit. As soon as the lamp has turned on, the output voltage is decreased to less than 20 V_{DC}.

The controller card defines the stabilized output current. Two ranges are possible to set; approximately 5 A and 8 A_{DC}. The lower range is appropriate for 75 watt lamps, while the higher one is used for 150 watt lamps. A jumper position on the controller card defines the current range. The two output currents can be adjusted individually on two trim potentiometers.

The output current can be checked using a standard voltmeter. At the test point on the power supply card, the output current is presented as a measuring voltage. The voltage is pre-set at the factory so that 1.00 V corresponds to an output current of 10.0 A.

The ignition unit contains a high frequency oscillator, similar to the ones in electronic flashes for photographical use. The unit stays passive as long as a lamp current is flowing through the cables. When no current is detected, and the output voltage from the power supply card is 80 V_{DC}, a high voltage pulse is generated within a few seconds. As long as the lamp remains off, a new high voltage pulse is generated approximately every fifth second. The pulse energy is less than 0.1 Ws.

Cable length between the power supply card and the ignition unit can be up to 100 m (300 ft).

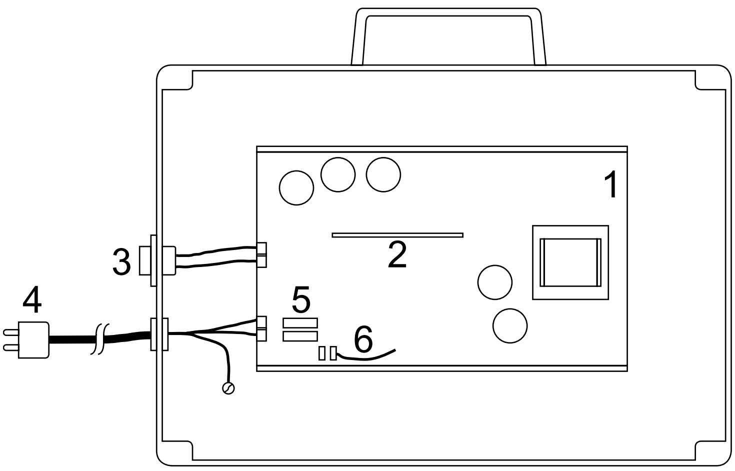


Figure 1.1. The PS150 unit.

Main parts of the PS150 unit	
1	Power supply card
2	Controller card
3	Voltage output
4	Mains input
5	Fuses
6	Jumper, 115/230 V mains voltage

1.3 The power supply card

The component layout of the power supply card is shown in figure 1.2.

The jumper 115/230 V_{AC} should be set according to the available input voltage. The fuses should have the following ratings:

Fuses	
230 V (+ 6 %, -10 %)	2 x 4 A / 250 V slow
115 VAC (± 10 %)	2 x 4 A / 250 V slow

At the test point, a voltage which is proportional to the output current can be measured. The voltage is adjusted at the factory so that 0 to 1 V_{DC} corresponds to 0 to 10 A. The adjustment is made on a trim potentiometer on the control card (TP adjust). However, it should not be moved from its original setting.

The output ratings are approximately 19 V_{DC}/8.0 A, or 14 V_{DC}/5.4 A, depending on the jumper setting on the controller card.

When the output current is zero, the output voltage is increased to about 80 V_{DC}. The voltage has two purposes. First, it supports the ignition of the lamp, and second, it activates the ignition unit to generate high voltage pulses for the ignition.

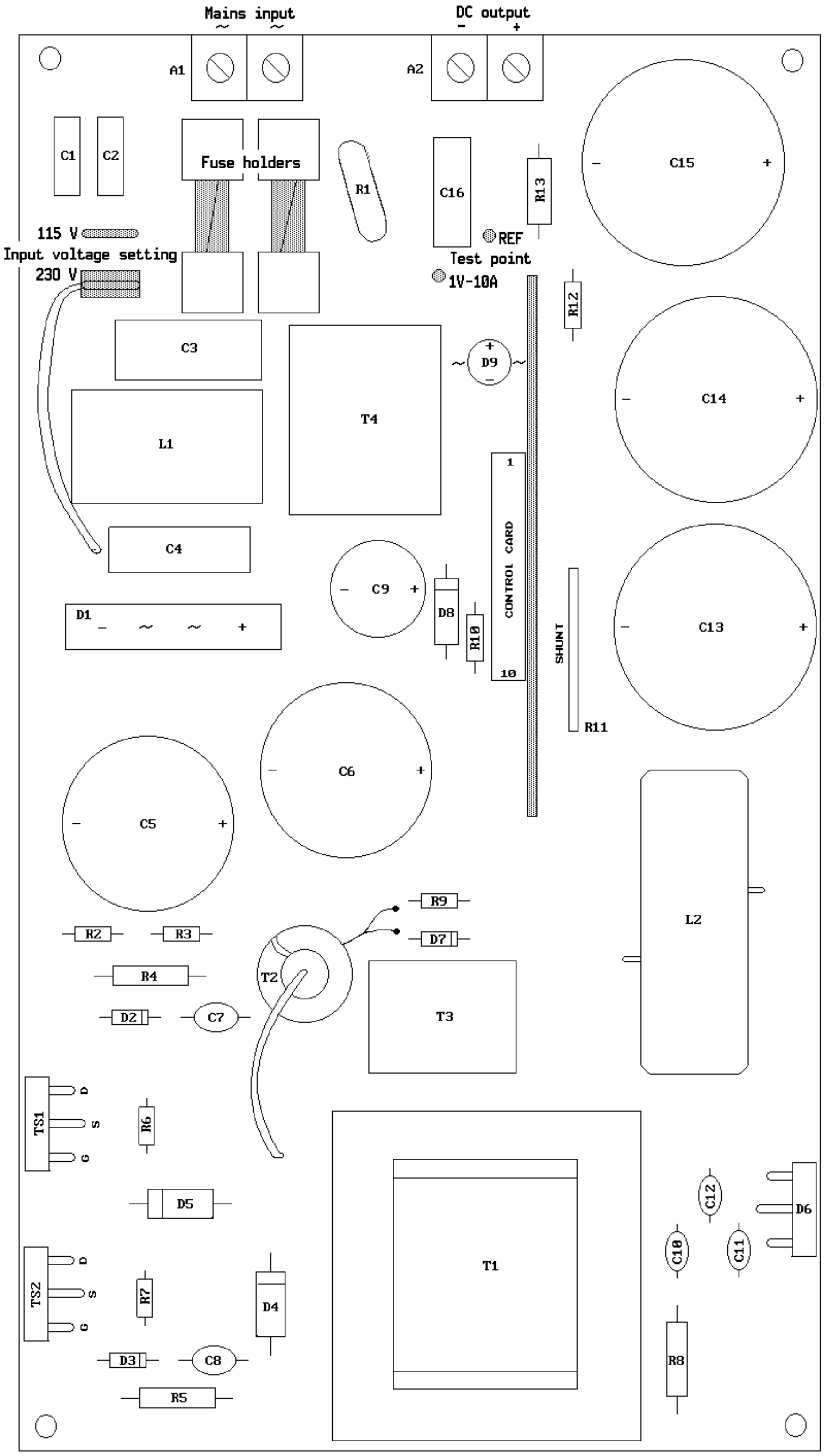


Figure 1.2. *The power supply card, component layout.*

1.4 The control card

The control card is mounted directly onto the power supply card. The component layout is shown in figure 1.3.

The card has three trim potentiometers. Two of them are used to adjust the output current for respective range. The third one was set at the factory so that the test point voltage 0 to 1 V corresponds to the output current 0 to 10 A. The last one should therefore not be touched.

The jumper is set according to the output requirements. The 4 A setting is used when the lamp rating is 75 W, while 8 A is consequently used for 150 W lamps.

The following lamp currents should be set for respective wattage:

Lamp current	
75 W	5.4 A (TP voltage 0.54 V)
150 W	8.0 A (TP voltage 0.80 V)

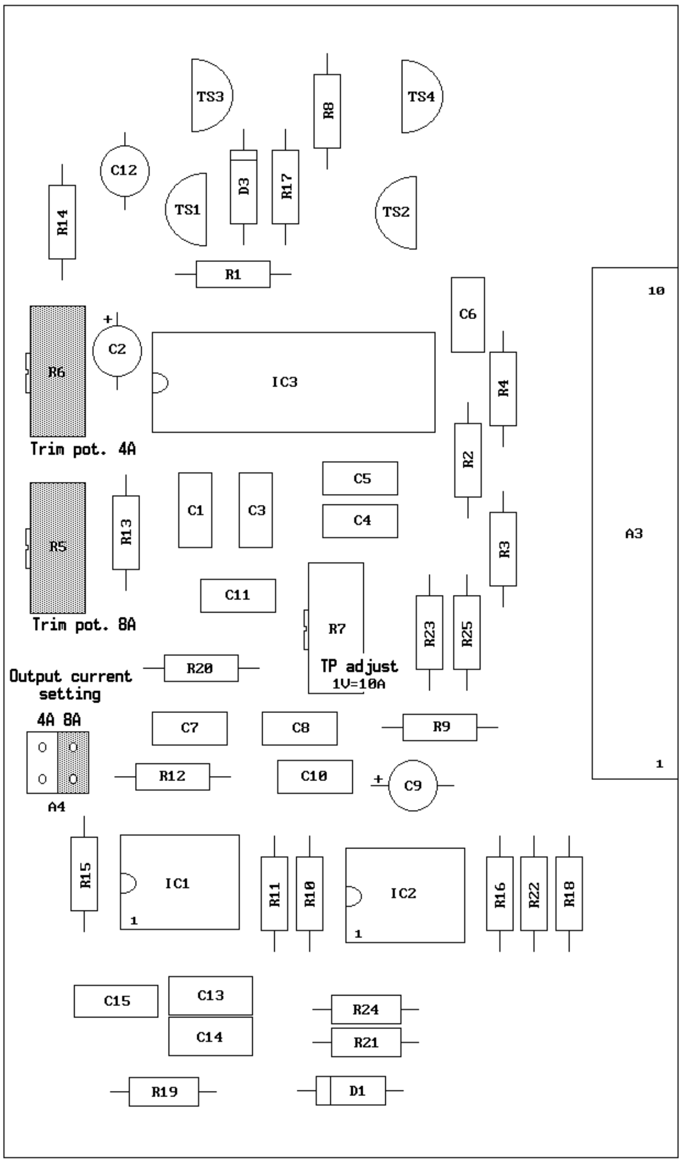


Figure 1.3. The control card. The lamp current is adjusted on the potentiometer corresponding to the jumper setting, 4 A respectively 8 A.

1.5 The ignition unit

The ignition unit is activated by the DC voltage of 80 V delivered by the power supply card when the lamp current is zero. As long as the voltage is on, a high voltage pulse is generated approximately every fifth second.

The pulse duration is very short, in the order of a few milliseconds. This means that, although the peak voltage is about 30 000 V, the pulse energy is less than 0.1 Ws.

The ignition unit is mounted directly onto the emitter. The cables through which the high voltage pulse is distributed are very short, and the disturbances on electronic equipment near the power supply are thus eliminated. The unit is then contained in a casing which has an input terminal for the supply voltage and two output terminals for the lamp cables.

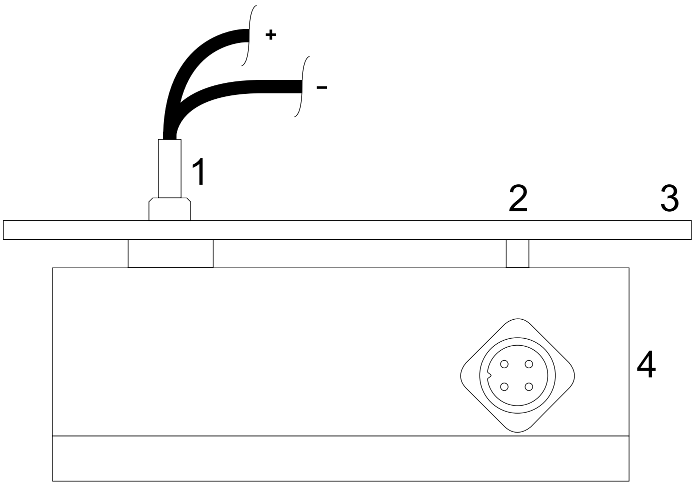


Figure 1.4. The ignition unit box, here attached to an emitter. The AMP connector is the input terminal.

Main parts of the ignition unit	
1	Lamp cables
2	Attachment
3	Emitter
4	AMP connector

1.6 The cable

The conductor between the power supply and the emitter/ignition unit is a three conductor cable (3 x 1.5 mm^2), terminated with AMP plugs. One conductor is used for plus, one for minus, while the last one is the earth (i.e. protective ground) connection.

Standard length is 5 metres, but can be extended up to 100 metres (15 to 300 ft).

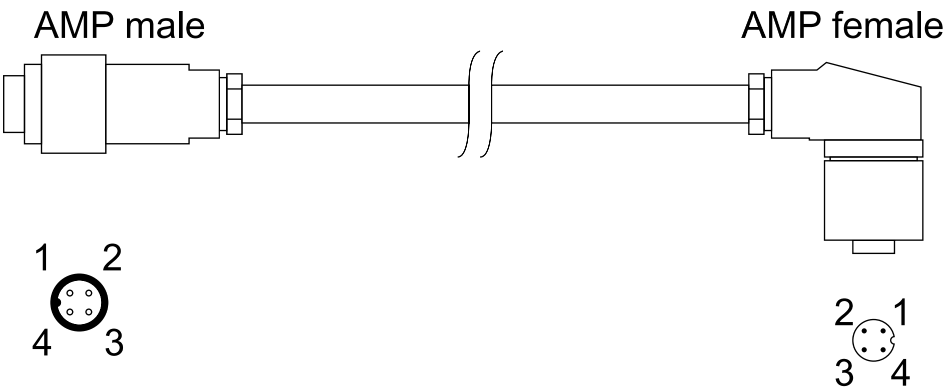
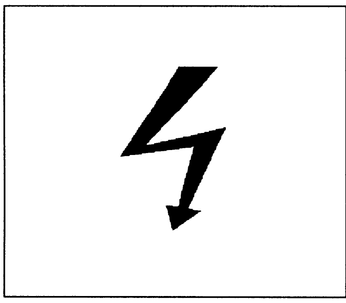
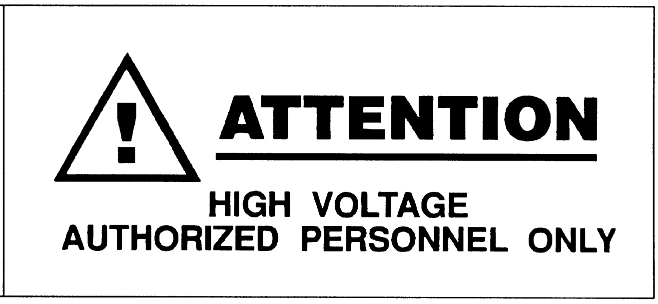


Figure 1.5. Cable between the power supply and the external ignition unit.

Conductors	
1	+
2	-
3	NC
4	Earth

Safety regulations 2





- These signs indicate that the power unit is a high voltage unit. Service should therefore be carried out only by personnel authorized by Opsis.
- Ensure that the unit is properly earthed via the mains cable.
- Also ensure that the emitter is earthed via the power supply.
- Use only original cables, and never join cables.
- Ensure that the power supply is set in accordance with the mains voltage (115/230 V_{AC}) before it is connected.

NOTE: *The voltage may be changed by authorized service personnel only.*

- Do not attempt to work on the unit or the emitter without first disconnecting the power unit from the mains.
- Before changing a lamp, or attempting any other work on the emitter, study the safety regulations in the emitter manual carefully. Protective glasses should be worn at all times when working with the lamp.

Installation and maintenance 3

3.1 Installation

1. When choosing the site for the power supply, avoid positioning the unit close to radiators, heating elements, or where it is exposed to direct sunlight.
2. When installing the unit, the safety regulations given in section 2 should be followed. Look inside the unit for loose or missing electrical connections or circuit boards, or broken items. No loose screws, bits of wire, dirt, moisture, etc., may be present.
3. Before the unit is connected to the power source:
   - Connect the cable between the power supply and the ignition unit.
   - Ensure that the input voltage jumper setting is in accordance with the mains voltage (115/230 V_{AC}).
   - Check that the correct fuses are securely in place in the fuse sockets (2 x 4 A/ 250 V for both 115 V_{AC} and 230 V_{AC}).
   - Ensure that the output current jumper setting is correct for the lamp type and wattage used (4 A for 75 W, and 8 A for 150 W).
4. Connect the power supply to the mains. If the lamp does not light, please refer to section 4, Troubleshooting.
5. The lamp current should be checked when the lamp has been on for at least 30 minutes (authorized personnel only). Use a voltmeter to measure at the test point on the power supply card (see section 1.3). Adjust the lamp current on the corresponding trim potentiometer on the control card to the correct measure voltage (0.54 V for 75 W lamps, and 0.80 V for 150 W lamps).

3.2 Preventive maintenance

Under normal conditions, the power supply requires little maintenance. However, it is recommened that the following precautions be taken once a month:

- Check inside the unit for possible water leakage or dust contamination, indications of overheating, etc.
- If the unit has been exposed to vibrations, ensure that no circuit board or electrical connection is loose.
- Check the measure voltage at the test point, and adjust if necessary.

Troubleshooting 4

This section provides guidance for determining the causes of malfunctions or anomalies during operation. The guidance is given in the form of a troubleshooting scheme, which may help to diagnose the problems.

Before attempting to work on the unit, carefully study the safety regulations given in section 2.

Problem	**Possible causes and action**	
No high voltage pulse is generated.	Check that main line is on.	Verify correct line voltage and frequency.
	Mains socket earthed.	Unit must have 3-wire safety power cord.
	Check the main fuses.	Replace fuses if ohm meter indicates open circuit.
	Check that power cable is properly connected.	Connect power cable properly.
	Check that output voltage from power supply card is 80 V.	Replace power supply card, if failed.
	Check that input voltage to ignition unit is 80 V.	Replace ignition unit, if failed.
	No damage on power cable.	Replace power cable, if damaged.
High voltage pulse is generated but lamp does not turn on.	Lamp is worn.	Replace lamp
	Connection to lamp anode is bad.	Disconnect wire and polish the terminal.
	The high voltage pulse is not reaching the lamp.	Check lamp cables for damages. Replace if necessary.
	Current supply has failed.	Replace the power supply card.
New lamp fails after a short period in use.	The lamp was faulty.	Replace with a new lamp.
	Incorrect current setting.	Set jumper on control card in accordance with lamp ratings.
	The lamp current was too high.	Adjust on control card. If not possible to adjust, replace control card.